

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R,S) / S4 (PT) (R,S) Examination May 2024 (2019 Scheme)

Course Code: CST304**Course Name: COMPUTER GRAPHICS AND IMAGE PROCESSING**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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| 1 | Write any three applications of computer graphics. | (3) |
| 2 | Find the points in the line from (5, 6) to (8, 12) using the DDA line drawing algorithm. | (3) |
| 3 | Given a triangle with vertices at coordinates (10,20), (10,10), (20,10). Find the coordinates of vertices after Scaling with parameters $S_x=2$, $S_y=1.5$, with respect to the origin. | (3) |
| 4 | List the steps for general pivot point rotation. | (3) |
| 5 | Differentiate parallel and perspective projections. | (3) |
| 6 | Perform window to viewport transformation for the point (20,15). Assume that (X_{wmin}, Y_{wmin}) is (0, 0); (X_{wmax}, Y_{wmax}) is (100, 100); (X_{vmin}, Y_{vmin}) is (5, 5); (X_{vmax}, Y_{vmax}) is (20, 20). | (3) |
| 7 | List any three applications of digital image processing. | (3) |
| 8 | The spatial resolution of an image is given by 128 X 128. What is the storage requirement if it is represented by 256 gray levels? | (3) |
| 9 | Explain the terms (i) Smoothing (ii) Sharpening | (3) |
| 10 | Explain any two thresholding techniques used in image segmentation. | (3) |

PART B

Answer one full question from each module, each carries 14 marks.

Module I

- 11 a) Explain any seven applications of computer graphics. (7)
- b) Write Bresenham's circle drawing algorithm. Find the points in a circle octant in the first quadrant with the centre point coordinates (0, 0) and radius as 8. (7)

OR

- 12 a) Explain the working of the raster scan system with suitable figures. (7)
- b) Explain Bresenham's Line drawing algorithm with the help of an example. (7)

Module II

- 13 a) Explain scan line polygon fill algorithm. What problem does the algorithm encounter when a scan line passes through a vertex? (7)
- b) Perform the following transformation on a triangle with vertices A(0, 0), B(1, 0) and C(1, 1). Find out the new coordinates and draw the result of each transformation. (7)
- i) Rotate the triangle by 90 degree anticlockwise direction.
- ii) Reflection about x axis.

OR

- 14 a) Explain Boundary fill algorithm to fill the interior of any specified area. Differentiate Boundary fill and flood fill algorithms. (7)
- b) Prove that the multiplication of transformation matrices for each of the following sequence of operations is commutative : (7)
- i) Any two successive translations.
- ii) Any two successive scaling operations.

Module III

- 15 a) Explain Cohen Sutherland line clipping algorithm with an example. (7)
- b) Explain the Scan line method for visible surface detection. (7)

OR

- 16 a) Explain Depth buffer algorithm for visible surface detection. Write any two disadvantages of the Depth buffer algorithm. (7)
- b) Describe perspective projection. Derive transformation matrix for perspective projection. (7)

Module IV

- 17 a) With the help of a block diagram, describe in detail the fundamental steps in image processing. (7)
- b) What is image convolution? Explain various steps involved in image convolution with the help of an example. (7)

OR

- 18 a) Explain the basic relationships between pixels? (7)
- b) Explain the concept of image sampling and quantization. (7)

Module V

- 19 a) What is image segmentation? Explain edge and region based segmentation technique. (7)
- b) Explain histogram equalization method of image enhancement. (7)

OR

- 20 a) Explain any two gray level transformation functions. (7)
- b) Describe how an image is segmented using split and merge technique in association with the region adjacency graph. (7)
